IV Semester B.Tech (Common to CS/IT/CCE)

Mid Term Examination March 2024

Subject: Engineering Mathematics IV (MAT 2226)

Date: 23 March 2024 Time: 10:45 am - 12:45 pm

Max. Marks:30

Type: MCQ Duration: 20 min.

Q1. If the probability of hitting the target is 0.5 for A, 0.25 for B, 0.75 for C then find the probability of at least one of them hits the target is…. (0.5)

1. $\frac{29}{32}$
2. $\frac{1}{32}$
3. $\frac{3}{32}$
4. $\frac{30}{32}$

Q2. A continuous random variable has pdf $f(x) = kx^2e^{-x}$; $x \geq 0$. Then the value of $k =$… (0.5)

1. $\frac{1}{2}$
2. 3
3. 2
4. $\frac{1}{4}$

Q3. Two independent random variables X and Y have variances 0.2 and 0.5 respectively. Let $Z = 5X - 2Y$. The variance of Z is…… (0.5)

1. 3
2. 7
3. 5
4. 4

Q4. Six fair coins are tossed. Then the probability of getting exactly 3 heads is….. (0.5)

1. $\frac{21}{32}$

2. $\frac{3}{32}$
3. $\frac{5}{16}$
4. $\frac{7}{16}$

Q5. If S and T are two independent events with $P(S) < P(T), P(S \cap T) = \frac{6}{25}$ and $P(S|T) + P(T|S) = 1$, then $P(S)$ is …….. (0.5)

1. $\frac{1}{5}$
2. $\frac{6}{25}$
3. $\frac{2}{5}$
4. $\frac{1}{25}$

Q6. The marks obtained were found normally distributed with mean 75 and variance 100. The percentage of students who scored more than 75 marks is ……….. (0.5)

1. 50
2. 25
3. 74
4. 100

Q7. Suppose a fair die is rolled 200 times. What is the expected number of rolls giving prime numbers? (0.5)

1. 50
2. 100
3. 150
4. 200

Q8. The cumulative distribution function of a random variable X is given by, $F(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & Otherwise \end{cases}$. Then the probability density function of X is ……. (0.5)

1. $f(x) = \begin{cases} x + \frac{e^{-2x}}{2}, & x \geq 0 \\ 0, & Otherwise \end{cases}$
2. $f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & Otherwise \end{cases}$
3. $f(x) = \begin{cases} 1 - 2e^{-2x}, & x \geq 0 \\ 0, & Otherwise \end{cases}$
4. $f(x) = \begin{cases} 1 + 2e^{-2x}, & x \geq 0 \\ 0, & Otherwise \end{cases}$

Q9. Covariance$(X+Y, X-Y) = \ldots\ldots\ldots$ (0.5)

1. $V(X)+V(Y)$
2. *V(X) – V(Y)*
3. $V(X^2)-V(Y^2)$
4. $V(X^2)+V(Y^2)$

Q10. Fifty tickets are serially numbered from 1 to 50. One ticket is drawn from these tickets at random. The probability of its being a multiple of 3 or 4 is………… (0.5)

1. $\frac{12}{25}$
2. $\frac{14}{25}$
3. $\frac{2}{5}$
4. $\frac{1}{5}$

Type: Descriptive Questions Time: 1 hr 40 min.

Q11. The probability of Tom hitting a target is 1/3.

(a) If Tom fires 5 times, what is the probability of his hitting the target at least twice?

(b) How many times must Tom fire so that the probability of his hitting the target at least once is more than 90%? .(4)

Q12. A Wall Street analyst estimates that the annual return from the stock of company A can be considered to be an observation from a normal distribution with mean $\mu = 8\%$ and standard deviation $\sigma = 1.5\%$. The analyst's investment choices are based upon the considerations that any return greater than 5% is "satisfactory" and a return greater than 10% is "excellent." Find the probability that company A's stock will prove to be "unsatisfactory." Find the probability that company A's stock will prove to be "excellent" .(4)

Q13. Consider the family of n children. Let A be the event that a family has children of both sexes. Let B be the event that there is at most one girl in the family. Find the value of n for which the event A and B are independent. Assume that each child has probability $\frac{1}{2}$ of being a boy.(3)

Q14. Box 1 contains 4 black and 5 green balls and Box 2 contains 5 black and 4 green balls. 3 balls are randomly drawn from box 1 without replacement and transferred to box 2 and then a ball is drawn from box 2 and is found to be green. What is the probability that 2 green and 1 black balls are transferred from box 1?.(3)

Q15. At a telephone centre, the time X(in minutes) for which an agent speaks on

a telephone is found to be random, for which the distribution function is given by

$$f(x) = \begin{cases} kx, & 0 \le x \le 2 \\ 2k, & 2 \le x \le 4 \\ k(6-x), & 4 \le x \le 6 \end{cases}$$

a. Find the value of k, for which $f(x)$ is valid.
b. Find cumulative distribution function $F(x)$.
c. Find $P(4 \le X < 5 \mid X > 3)$.(3)

Q16. Let (X, Y) be a two-dimensional random variable with the joint pdf

$$f(x,y) = \begin{cases} kxy & 0 \le x \le 1, 0 \le y \le \sqrt{x} \\ 0 & elsewhere. \end{cases}$$

Find the constant k and marginal pdfs of X, Y .(3)

Q17. If X_1, X_2, X_3 be uncorrelated random variables having same standard deviation. Find the correlation coefficient between $U = X_1 - X_2$ and $V = X_3 + X_2$.(3)

Q18. Assuming that the year has 365 days, what is the probability that in a random group of k people at least two people share the same birthday? .(2)